

04 Airflow Workflow End-to-End Execution Guide

Airflow Workflow End-to-End Execution Guide

Overview

This guide covers the complete end-to-end execution of an Airflow workflow, from post-setup configuration to successful execution and validation. This assumes you have already created your MWAA environment.

Prerequisites

- ✓ MWAA Environment created and running
- ✓ Redshift Serverless environment configured
- ✓ S3 buckets prepared
- ✓ DAG script ready with correct bucket names

Step-by-Step Execution

1. Configure IAM Role Permissions

Add S3 Full Access to Airflow Role

1. Navigate to your **MWAA Environment**
2. Scroll down to **Permissions** section
3. Click on the **IAM Role/Execution Role** link
4. In IAM console, click "Add permissions" → "Attach policies"
5. Search for "S3"
6. Select "AmazonS3FullAccess"
7. Click "Add permissions"

Why this is needed: Airflow has default access to its DAGs bucket, but needs access to data input buckets for CSV files.

2. Prepare Data Buckets

Create Data S3 Bucket Structure

1. **Create new S3 bucket** (different from Airflow DAGs bucket)
2. **Create folder structure:**

```
your-data-bucket/  
├─ spotify-data/  
│   ├─ songs.csv  
│   ├─ users.csv  
│   └─ streams/  
│       └─ streams_1.csv (or streams_2.csv, streams_3.csv)
```

Upload Required Files

1. Inside `spotify-data/`:

- Upload `songs.csv`
- Upload `users.csv`

2. Inside `spotify-data/streams/`:

- Upload one of the stream files (`streams_1.csv` , `streams_2.csv` , or `streams_3.csv`)

3. Upload DAG Script

Prepare and Upload DAG

1. **Before uploading:** Update bucket names in your DAG script to match your bucket names
2. **Navigate to Airflow DAGs S3 bucket**
3. **Go to `dags/` folder**
4. **Drag and drop** your Airflow DAG script
5. ⌚ **Wait 2-3 minutes** for DAG to appear in Airflow UI

4. Configure Redshift Connection

Get Redshift Endpoint

1. Navigate to **Amazon Redshift console**
2. Click on your **workgroup**
3. **Copy the endpoint** (click copy icon)
4. **Remove** database name and port from copied endpoint

Create Airflow Connection

1. **Access Airflow UI** from MWAA environment
2. **Admin → Connections**
3. **Click "+" (plus icon)**
4. **Configure connection:**
 - **Connection ID:** `redshift_default`
 - **Connection Type:** `Redshift`
 - **Host:** `[your-redshift-endpoint]` (without database and port)
 - **Schema:** `songs_db`
 - **Login:** `admin`
 - **Password:** `[your-redshift-password]`
 - **Port:** `5439`
5. **Click "Save"**

Critical: Connection ID must match exactly what's used in your DAG script (`redshift_default`)

5. Execute the Workflow

Start DAG Execution

1. **Navigate to DAGs** in Airflow UI
2. **Verify your DAG** is visible (may take 2-3 minutes)
3. **Click the play button** (▶) to trigger execution
4. **Click on DAG name** to view details
5. **Select "Graph"** view to see workflow structure

Monitor Execution

- **Expected flow:**

```
Data Validation → Branch (Pass/Fail) → Data Transformation → Move Processed Files
```

- **Execution time:** ~2-3 minutes
- **Success indicator:** All tasks turn green

6. Workflow Validation

Check XCom Data Transfer

1. **Click on first task** in the graph
2. **Select "XCom"** tab
3. **Verify:** Validation results dictionary is visible
4. **Confirms:** Data passed successfully between tasks

Check Logs for Debugging

1. **Click on any task**
2. **Select "Logs"** tab
3. **Review:** Execution statements and results
4. **Use for troubleshooting** if needed

7. Validate Results

Verify Redshift Data

1. **Navigate to Redshift Query Editor**
2. **Run queries to check data:**

```
SELECT * FROM genre_level_kpis LIMIT 10; SELECT * FROM monthly_kpis LIMIT 10;
```

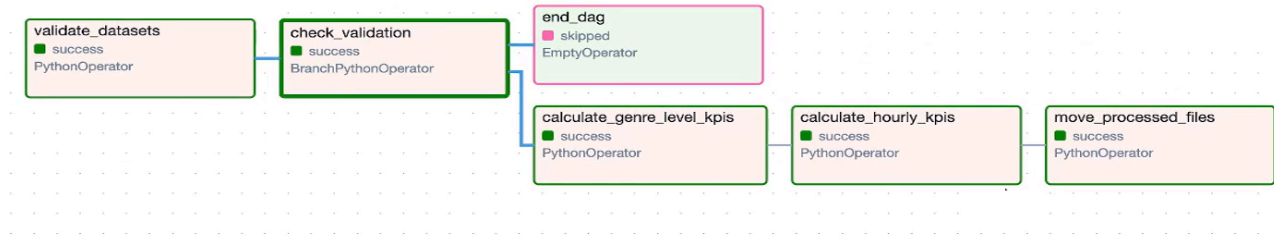
3. **Verify:** Data is populated correctly

Check File Archival

1. **Navigate to data S3 bucket**

2. **Go to:** `spotify-data/streams/`
3. **Verify:** `archived/` folder created
4. **Check:** Stream file moved to archived folder

Expected Workflow Structure



Troubleshooting Guide

Common Issues and Solutions

DAG Not Appearing

- **Wait:** 2-3 minutes for sync
- **Check:** S3 bucket path in MWAA configuration
- **Verify:** DAG syntax is correct

Connection Failed

- **Check:** Redshift endpoint format (no database/port)
- **Verify:** Redshift publicly accessible enabled
- **Confirm:** Security group allows port 5439

Permission Errors

- **Verify:** S3FullAccess policy attached
- **Check:** Bucket policies allow access
- **Confirm:** IAM role has correct permissions

Task Failures

- **Review:** CloudWatch logs
- **Check:** Task logs in Airflow UI
- **Verify:** Data file formats and locations

Debugging Best Practices

1. **Always check logs** for detailed error messages
2. **Use XCom** to verify data transfer between tasks
3. **Monitor CloudWatch** for environment-level issues
4. **Test connections** before running full workflow

Success Criteria Checklist

- ☐ DAG appears in Airflow UI
- ☐ All tasks execute successfully (green status)
- ☐ Data appears in Redshift tables
- ☐ Files are archived in S3

- ☐ XCom shows data transfer between tasks
- ☐ No errors in logs

Post-Execution Tasks

Cleanup (Optional)

1. **Pause DAG** if not needed for regular execution
2. **Monitor costs** in AWS Cost Explorer
3. **Archive logs** if needed for compliance

Next Steps

1. **Create additional DAGs** for other workflows
2. **Set up scheduling** for regular execution
3. **Implement monitoring** and alerting
4. **Optimize performance** based on execution metrics

Key Learning Points

Workflow Components

- **Data Validation:** Ensures data quality before processing
- **Branching:** Handles conditional logic based on validation
- **Data Transformation:** Processes and loads data to warehouse
- **File Management:** Archives processed files

Integration Points

- **S3:** Data storage and DAG hosting
- **Redshift:** Data warehouse for analytics
- **Airflow:** Workflow orchestration
- **IAM:** Security and permissions management

Resources

- [Airflow XCom Documentation](#)
- [MWAA Troubleshooting Guide](#)
- [Redshift Connection Guide](#)

 Congratulations on completing your first Airflow workflow!